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Study of the pharmacokinetics of various drugs under conditions of antiorthostatic hypokinesia and the pharmacokinetics of acetaminophen under long-term spaceflight conditions

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Abstract

Objectives: To study the pharmacokinetics and relative bioavailability of drugs of different chemical structure and pharmacological action under conditions simulating the effects of some factors of spaceflight, as well as the peculiarities of the pharmacokinetics of acetaminophen under long-term spaceflight conditions.

Methods: The pharmacokinetics of verapamil (n=8), propranolol (n=8), etacizine (n=9), furosemide (n=6), and acetaminophen (n=7) in healthy volunteers after a single oral administration under normal conditions (background) and under antiorthostatic hypokinesia (ANOH), the pharmacokinetics of acetaminophen in spaceflight members under normal ground conditions (background) (n=8) and under prolonged spaceflight conditions (SF) (n=5) were studied.

Results: The stay of volunteers under antiorthostatic hypokinesia had different effects on the pharmacokinetics and bioavailability of drugs: Compared to background, there was a decreasing trend in V_z for verapamil (−54 Δ%), furosemide (−20 Δ%), propranolol (−8 Δ%), and acetaminophen (−9 Δ%), but a statistically significant increase in V_z was found for etacizine (+39 Δ%); there was an increasing trend in Cl_t for propranolol (+13 Δ%) and acetaminophen (+16 Δ%), and a decreasing trend in Cl_t for etacizine, verapamil, and furosemide (−22, −23 and −9 Δ% respectively) in ANOH. The relative bioavailability of etacizine, verapamil, and furosemide in ANOH increased compared to background (+40, +23 and +13 Δ%, respectively), propranolol and acetaminophen decreased (−5 and −12 Δ% accordingly). The relative rate of absorption of etacizine and furosemide in ANOH decreased (−19 and −20 Δ%, respectively) while that of verapamil, propranolol, and acetaminophen increased (+42, +58 and +26 Δ%, respectively). A statistically significant decrease in $AUC_{0\text{-}\infty}$ (−57 Δ%), C_{max} (−53 Δ%), relative bioavailability of acetaminophen (−52 Δ%) and a sharp increase in Cl_t (+147 Δ%), T_{max} (+131 Δ%) as well as a trend towards a significant decrease in $T_{1/2}$ (−53 Δ%), MRT (−36 Δ%) and a moderate increase in V_z (+24 Δ%) were found under control compared to background. Unidirectional changes in $AUC_{0\text{-}\infty}$, Cl_t, $T_{1/2}$, MRT and relative bioavailability of acetaminophen, which are more pronounced in SF and opposite dynamics for C_{max}, T_{max}, V_z were found in ANOH and SP compared to background studies.

Conclusions: The data obtained allow recommending the studied drugs for rational pharmacotherapy in the possible development of cardiovascular disease in manned spaceflight.

Keywords: acetaminophen; antiorthostatic hypokinesia; etacizine; furosemide; pharmacokinetics; propranolol; spaceflight; verapamil.

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Introduction

Manned missions to deep space are currently planned [1–5], primarily to Mars [3, 5–10], which may take place in the next 10–20 years [3, 8], and their duration may be 900–1,100 days [8]. Such long reconnaissance missions